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Evaluating a Digital Serious Game for Learning Medical Terminology in a Randomized Controlled Trial

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Abstract

Healthcare students often struggle with learning medical terminology due to its complexity and abstract nature. This randomized controlled trial assessed the effectiveness of *MedQuiz*, a digital serious game, in enhancing immediate terminology acquisition and user satisfaction among 60 undergraduate students in health-related programs. Participants were assigned to either a control group with traditional instruction or an intervention group using *MedQuiz* alongside lectures. Post-test scores were significantly higher in the intervention group ($P < .001$), and user experience predicted performance. Usability metrics (SUS = 90.36%) and playability ratings indicated strong engagement. The MEEGA+ (Metrics for Educational Game Assessment+) framework showed positive perceptions of usability and learning, while entertainment value was moderate. Findings support *MedQuiz* as a scalable and engaging tool for short-term medical terminology learning. Long-term retention was not assessed. Further studies should examine delayed learning outcomes and explore integration with artificial intelligence-based spaced repetition systems for sustained knowledge acquisition.

Protocol and ethics approval: The study protocol was published (DOI: 10.3233/SHTI220658). The study was approved by the Ethics Committee of Mashhad University of Medical Sciences (Approval No.: IR.MUMS.REC.1400.336). The study was not registered in a clinical trial registry.
Keywords: Digital serious game; medical terminology; gamified learning; medical education; randomized controlled trial

Introduction

The continuous evolution of healthcare demands innovative teaching methods to train competent medical professionals capable of adapting to the dynamic needs of society. Traditional medical education has primarily relied on passive learning methods, such as lectures, textbooks, and

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flashcards, which, although effective in delivering information, have significant limitations in fostering long-term knowledge retention and active engagement [1]. This challenge is particularly evident in medical terminology education, where students must memorize many complex terms, often leading to rote memorization, a lack of profound comprehension, and difficulties in real-world application [2].

Research indicates that more than 50% of medical terminology knowledge is forgotten within a year due to limited interactive learning opportunities and the absence of repetition-based engagement [3, 4]. This knowledge decay is exacerbated by time constraints in curricula, reduced classroom hours, and the lack of active recall methodologies [5]. Additionally, proficiency in medical terminology is essential for clear communication among healthcare providers, accurate clinical documentation, and ultimately, patient safety [6]. The inability to effectively grasp and retain medical terminology can result in miscommunication, misinterpretation of clinical data, and medical errors, emphasizing the need for more engaging, retention-focused, and evidence-based educational strategies [7].

Digital serious games have emerged as a promising approach to address these challenges, offering an interactive, engaging, and student-centered method for learning [8]. Digital serious games integrate cognitive psychology principles, such as Flow Theory and Self-Determination Theory (SDT), to create an immersive learning environment where students actively engage in learning rather than passively consuming information [9].

Several studies have demonstrated that gamified learning enhances engagement, improves cognitive skills, increases knowledge retention, and facilitates real-world skill application [10-18]. A study by Seidlein et al. (2020) on the TERMInator gamified tool for medical terminology education found that digital serious games significantly improved students' knowledge and engagement compared to traditional teaching methods [3]. Similarly, Felszeghy et al. (2019) reported that incorporating digital serious games in anatomy education significantly enhanced student performance and motivation [15].

Additionally, a recent large-scale study conducted by Sarbaz et al. (2023) at Mashhad University of Medical Sciences (MUMS) evaluated the impact of digital serious games in medical education among 496 medical and paramedical students. Their findings indicated that over 80% of students considered digital serious games an effective tool for learning, improving motivation, and enhancing educational experiences. The study highlighted interactive features such as competition, scoring mechanisms, and adaptive challenges, reinforcing the potential of digital serious games as an innovative learning strategy in medical education [16]. However, while many previous studies have primarily focused on short-term learning gains, recent longitudinal research has provided compelling evidence that digital serious games can significantly enhance long-term knowledge retention among healthcare students. Furthermore, a meta-analysis by Lee et al. [19] confirmed that digital serious games in nursing education can enhance knowledge and performance, particularly when integrated into structured learning periods. These findings underscore the potential of digital serious games not only for immediate learning, but also for supporting long-term memory consolidation in healthcare education.

Despite these promising findings, most existing digital serious games in medical education remain predominantly individual-focused and lack features that enable peer-based collaboration and comprehensive performance analytics [20]. Furthermore, previous studies have primarily focused on short-term knowledge acquisition rather than long-term retention and practical application in clinical settings [21]. Addressing these limitations requires a game-based educational intervention

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that facilitates both individual mastery and interactive peer learning while providing actionable feedback for students and educators [19, 22, 23].

To bridge this research gap, we developed MedQuiz, an innovative digital serious game tool designed to enhance medical terminology acquisition through interactive challenges, peer-based competition, and real-time, data-driven feedback. Unlike previous gamified learning tools, MedQuiz incorporates a multiplayer mode that enables students to compete with their peers, fostering social learning and reinforcing medical terminology concepts through interactive gameplay. Furthermore, MedQuiz features an advanced performance analytics system that provides students with personalized reports on their strengths and weaknesses across different medical terminology categories. This adaptive feedback mechanism allows students to focus on areas requiring improvement, while also equipping educators with valuable insights into student performance trends, helping them refine their teaching strategies for more effective learning outcomes.

To evaluate the effectiveness of this approach, we conducted a randomized controlled trial (RCT) to assess MedQuiz as a tool for medical terminology learning. We hypothesize that students in the MedQuiz group will demonstrate superior immediate learning performance, along with higher levels of cognitive engagement and self-efficacy compared to those receiving traditional instruction.

By providing empirical evidence on the impact of digital serious games in medical terminology education, this study aims to contribute to the growing body of research on digital innovations in medical education and inform the development of future technology-enhanced learning strategies in the health sciences. While previous studies have explored both short-term and long-term learning effects of digital serious games, the current study focuses solely on immediate learning outcomes. Long-term retention was not assessed in our design. This distinction is critical, as our findings should be interpreted within the scope of immediate post-intervention performance rather than sustained knowledge over time.

The objective of this study was to evaluate the effectiveness of MedQuiz, a complementary digital serious game, in enhancing medical terminology learning among healthcare students. Specifically, we compared its impact on knowledge acquisition, user experience (UX), and student engagement when used alongside traditional lecture-based instruction.

Results

Participant Flow and Baseline Characteristics

This RCT involved 60 undergraduate students from MUMS, enrolled in the Health Information Technology (HIT) and Speech Therapy programs. All eligible participants consented to participate in the study and were randomly allocated into two equal groups: the intervention group (30 participants) and the control group (30 participants). Notably, there were no dropouts throughout the study, and all participants completed the entire study protocol. The participant flow, including enrollment, randomization, allocation, and analysis phases, is depicted in the consolidated standards of reporting trials (CONSORT) flowchart (**Figure 1**). No participants were lost at any stage of the study. The completed CONSORT-eHEALTH checklist (v1.6.1) is provided in **Supplementary Table 1**.

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The demographic characteristics of participants in the intervention and control groups are presented in **Table 1.** No significant differences were observed between the groups in terms of gender, age, academic grade point averages (GPA), or field of study (P > 0.05), indicating baseline homogeneity prior to the intervention.

Table 1. Baseline demographic information of participants in the RCT

Variable	Total	Game group	Control group	P-value
Gender n (%)	Male: 23 (38.1) Female: 37 (67.1)	Male: 10 (33.3) Female: 20 (66.7)	Male: 13 (43.3) Female: 17 (56.7)	0.426
GPA (Mean ±SD)	17.7 ± 1.4	17.10 ± 1.55	17.05 ± 1.72	0.627
Age (Mean ± SD)	21.52 ± 1.56	21.93 ± 1.57	21.10 ± 1.47	0.688
Field of Study n (%)	HIT: 41 (68.3) Speech therapy: 19 (31.7)	HIT: 23 (76.7) Speech therapy: 7 (23.3)	HIT: 18 (60) Speech therapy: 12 (40)	0.682
Total n (%)	60 (100)	30	30	

Note: Age SD is narrow because participants were primarily second-semester HIT or fifth-semester (Speech Therapy) students, typically within 1–2 years of age. Age range: 18–27 years.

Pre-Intervention Game Playability and Usability Evaluation

Prior to the intervention, MedQuiz's playability was assessed using the heuristic evaluation of playability (HEP) framework. A panel of 12 experts, including professionals in medical education, health informatics, social medicine, biostatistics, game design, and student representatives, evaluated key aspects such as engagement, competitiveness, feedback mechanisms, and user-friendliness.

Interactive features such as peer-versus-peer competition and the ability to play against randomly selected opponents were highly praised for their impact on student engagement. Participants also found the performance history feature—allowing students to monitor their progress across different body systems—particularly useful for self-assessment and promoting active involvement in learning. This dimension received a high average rating (Mean = 4.65/5). While experts appreciated the feature, some recommended adding more visual diversity to enhance its appeal and functionality.

The competitiveness aspect of the game, especially the inclusion of a dynamic leaderboard and visible peer rankings, was identified as a strong motivator. These elements were seen to encourage repeated participation and foster a sense of healthy competition among students. The mean score for this component was 4.83 out of 5, reflecting its perceived effectiveness. Expert suggestions included incorporating celebratory visual and auditory cues to reward top performers and increase emotional investment.

The feedback and support systems in *MedQuiz* were also evaluated favorably. Features such as coin-based elimination of incorrect options and the immediate display of correct answers were considered effective in reinforcing learning and increasing user engagement (Mean = 4.75/5).

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However, experts advised integrating more detailed, topic-specific feedback—categorized by body systems—to help students pinpoint and address their learning deficiencies.

The user interface (UI) was described as simple, intuitive, and easy to navigate, contributing to an accessible learning experience (Mean = 4.24/5). Nevertheless, minor aesthetic improvements were proposed to further refine the design, including adjustments to the color palette and visual clarity of specific graphic elements.

In terms of accessibility and reusability, the option to compete against a system-controlled opponent was appreciated. It offered a flexible opportunity for repeated practice, particularly when peer opponents were unavailable. This component scored 4.53 out of 5. Expert recommendations focused on implementing adjustable difficulty levels for solo matches to accommodate different learner needs and skill levels.

Following the playability assessment, a System Usability Scale (SUS) evaluation was conducted to assess the overall UX. A separate group of seven experts in medical education, instructional design, and health informatics reviewed the system. The SUS yielded an overall score of 90.36, which indicates excellent usability and a high degree of user satisfaction. The results, presented in **Table 2**, show that participants found the system easy to use, intuitive, and well-integrated. Experts particularly noted the confidence users felt while navigating the system and the minimal need for technical support.
Despite the strong usability performance, some reviewers recommended simplifying the onboarding process and refining the tutorial section to better assist new users in quickly understanding game mechanics. These suggestions aim to enhance the early UX and reduce initial learning curves.

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Table 2. Mean SUS scores for each item (out of 5)

Question	Mean	SD
I think that I would like to use this system frequently	5.00	0
I found the system unnecessarily complex	1.00	0
I thought the system was easy to use	5.00	0
I think that I would need the support of a technical person to be able to use this system	1.00	0
I found the various functions in this system were well integrated	3.71	1.60
I thought there was too much inconsistency in this system	1.42	0.53
Would imagine that most people would learn to use this system very quickly	4.57	0.53
I found the system very cumbersome to use	1.00	0
I felt very confident using the system	4.57	0.53
I needed to learn a lot of things before I could get going with this system	2.28	1.60

Effect of the Intervention on Medical Terminology Knowledge

To assess the primary outcome—improvement in medical terminology knowledge—a one-way ANCOVA was conducted after confirming the statistical assumptions of normality, homogeneity of variances, and regression slopes. Levene's test for homogeneity of variances was non-significant ($F = 1.180$, $p = .282$), indicating equal variances across groups. The Shapiro-Wilk test for normality of residuals was non-significant ($W = 0.957$, $p = .064$), indicating that the assumption of normality was met. Additionally, skewness (0.391) and kurtosis (-0.741) values were within acceptable ranges. Given the sample size ($N = 60$), ANCOVA results were considered robust to minor deviations from normality.

Homogeneity of regression slopes was assessed by including a Group × pre-test score interaction term in the model. The interaction was not statistically significant ($F (1, 56) = 2.632$, $p = .110$), indicating that the relationship between pre-test and post-test scores did not differ significantly across groups. Therefore, the assumption of homogeneity of regression slopes was satisfied.

Post-test scores from the Medical Terminology Assessment demonstrated a statistically significant improvement in the intervention group compared to the control group ($P = 0.001$). ANCOVA analysis confirmed that the observed difference in post-test performance was statistically significant, indicating the effectiveness of the MedQuiz, digital serious game in enhancing medical terminology learning (**Figure 2**).

To quantify the magnitude of the intervention effect, Cohen's d was calculated using the pooled standard deviation of the post-test scores. The effect size was $d = 0.90$, indicating a large effect according to conventional thresholds. This supports the practical significance of the observed difference between groups beyond statistical significance ($P < .001$).

To examine the factors influencing post-test performance among students who used MedQuiz, a digital serious game, a multiple linear regression analysis was conducted with post-test score as the dependent variable. Independent variables included pre-test score, GPA, gender, field of study,

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usability perception, player experience, and the frequency of playing both digital and non-digital games (**Table 3**). Field of study was entered as a binary categorical variable, coded as 1 = HIT and 2 = Speech Therapy. Accordingly, Speech Therapy served as the reference group in the regression model.

The overall model was statistically significant ($F(8, 21) = 7.591$, $p < .001$), accounting for 74.3% of the variance in post-test scores ($R^2 = .743$, Adjusted $R^2 = .645$). The overall multiple regression model showed a large effect size (Cohen's $f^2 = 2.89$), indicating strong practical significance in explaining post-test score variance among participants in the intervention group. A detailed breakdown of effect sizes for individual predictors is provided in **Supplementary Table 2 and 3**.

Among the predictors, player experience ($B = 8.157$, $p = .006$) had the strongest influence, with a large unique effect ($sr^2 = 0.283$). Although field of study also reached statistical significance ($B = -10.066$, $p = .004$), its unique contribution was relatively small ($sr^2 = 0.002$), suggesting it may share variance with other predictors.

Other variables—including pre-test score ($sr^2 = 0.057$, $p = .153$), GPA ($sr^2 < 0.03$, $p = .222$), gender ($sr^2 = 0.031$, $p = .540$), usability perception ($sr^2 < 0.03$, $p = .238$), and both digital ($p = .522$) and non-digital ($p = .426$) game-playing frequency—did not significantly predict post-test performance. All exhibited small or negligible effect sizes ($sr^2 < 0.06$), indicating limited independent contribution to learning outcomes. These findings reinforce the central predictive role of player experience in digital game–based education.

Multicollinearity was examined using Variance Inflation Factors (VIF), with all predictors falling well below the standard threshold of concern (VIF < 1.34). This confirmed that the regression model was not compromised by multicollinearity. Specifically, VIF values were as follows: GPA (1.234), pre-test score (1.266), field of study (1.179), gender (1.178), digital game frequency (1.297), non-digital game frequency (1.339), usability score (1.209), and player experience score (1.272).

Given the modest sample size relative to the number of predictors, the multivariable regression results should be interpreted with caution due to potential overfitting. Multicollinearity was low (all VIFs < 2). Because an independent validation sample was not available and the study was not designed for predictive modeling, formal out-of-sample validation was not performed. Replication in larger cohorts is needed to confirm the stability and generalizability of these associations. The regression analysis was re-run on the intervention subgroup ($n = 27$), yielding a statistically significant model ($F(8,17) = 2.816$, $p = .032$) that explained 59.9% of the variance in post-test performance ($R^2 = .599$, Adjusted $R^2 = .386$). Although this represents a modest decrease compared to the full-sample model ($R^2 = .743$), these findings should be interpreted cautiously and are presented solely as an exploratory robustness check rather than evidence of model validation.

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Table 3. Result of regression analysis: factors affecting post-test scores

Predictor Variable	B (Unstandardized)	SE B	β (Standardized)	t	p-value
(Constant)	16.099	18.812	—	0.856	0.402
Pre-test Score	0.276	0.186	.193	1.483	0.153
GPA	1.071	0.850	.190	1.260	0.222
Gender	-1.434	2.305	-.079	-0.622	0.540
Field of Study	-10.066	3.064	-.495	-3.286	0.004
Frequency of digital game playing	0.743	1.140	.095	0.652	0.522
Frequency of non-digital game playing	0.984	1.212	.120	0.812	0.426
Usability Score	-7.140	5.882	-.174	-1.214	0.238
Player Experience Score	8.157	2.692	.414	3.030	0.006

Abbreviations: B, unstandardized coefficient; SE, standard error; β, standardized coefficient. Statistical significance was set at $p < .05$. Model statistics: $R^2 = 0.743$, Adjusted $R^2 = 0.645$, $F(8, 21) = 7.591$, $p < .001$

To assess the robustness of our findings, a sensitivity analysis was conducted using only the 20-item Medical Terminology Assessment specified in the original published protocol [24]. Although the assessment was expanded to 40 items prior to implementation—following curriculum mapping and expert panel recommendations to ensure comprehensive content coverage across all major body systems—this post-hoc analysis aimed to verify whether the observed effects were dependent on the additional items. The full question list is available in **Supplementary Table 4**.

The primary ANCOVA and regression analyses were re-run using only the original 20 items. Results remained statistically significant and consistent with those derived from the full 40-item assessment, indicating that the effectiveness of the MedQuiz intervention was not contingent upon the expanded item pool. Full details of the sensitivity analysis are presented in **Supplementary Table 5**.

Beyond learning outcomes, secondary analyses were conducted to investigate user engagement levels, gameplay patterns, and overall experience with the MedQuiz platform. All gameplay data—including time spent using the digital serious game, number of competition sessions, accuracy rates, and total scores—were automatically captured in the system database. Analysis of these engagement metrics showed that students in the intervention group used MedQuiz for an average of 3.53 hours per week and participated in approximately 7.43 competitive sessions per week. These findings reflect a high level of sustained user engagement. Subgroup analysis of gameplay logs revealed no statistically significant gender-based differences in usage patterns. Both male and female students in the intervention group demonstrated comparable weekly gameplay durations ($d = 0.14$, $p = 0.711$) and similar numbers of competition sessions ($d = 0.19$, $p = 0.684$), indicating that MedQuiz design fostered equitable engagement across genders. There was no prescribed or mandatory duration for game use. Students in the intervention group were free to engage with the platform at their discretion, and the platform served purely as an optional enhancement to the existing curriculum.

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To investigate whether sustained engagement influenced learning outcomes, participants were categorized into two groups based on their gameplay patterns:
Consistent players (n = 16): average weekly quiz participation ≥ 6 sessions
Sporadic players (n = 14): average weekly participation < 6 sessions
An independent samples ANOVA revealed that students in the consistent engagement group had significantly higher post-test scores (M = 21.2, SD = 1.1) compared to those with sporadic engagement (M = 19.7, SD = 1.4),
$F(1, 28) = 6.47$, $p = 0.017$, $\eta^2 = 0.19$, indicating a moderate effect size. This suggests that consistent gameplay over the 8-week period was associated with better knowledge acquisition, providing preliminary support for the role of repeated exposure in vocabulary consolidation.

To further quantify the relationship between engagement intensity and learning outcomes, Pearson correlation coefficients were computed. The total number of games played ($r = 0.726$, $p < .001$) and the number of hours spent using MedQuiz ($r = 0.715$, $p < .001$) were both significantly and positively correlated with post-test performance, indicating that higher engagement levels were associated with greater knowledge gains.
These findings demonstrate large effect sizes based on Cohen's guidelines ($r > .50$), providing strong evidence for the pedagogical relevance of sustained and repeated gameplay. Moreover, the strong correlation between hours and game frequency ($r = 0.840$) further validates the consistency of user engagement throughout the intervention period.

To evaluate UX and the perceived educational impact of the digital serious game, the Metrics for Educational Game Assessment+ (MEEGA+) questionnaire was administered. As shown in **Table 4**, the mean usability score was 3.44 out of 4, indicating a strong overall performance. The highest rating was recorded in the "Learning" category (3.56), confirming that the serious game effectively supported the acquisition of medical terminology.

In terms of UX, the top scores were observed in the domains of "Confidence" (3.25) and "Focused Attention" (3.03), suggesting that the game's design successfully enhanced learners' self-assurance and sustained concentration during gameplay. In contrast, the "Entertainment" category received a comparatively lower score (2.89), indicating that future versions of the game could benefit from integrating additional serious game elements to further boost engagement and enjoyment. Despite the moderate entertainment score, further analysis revealed that this dimension played a meaningful role in sustaining user engagement. Specifically, Pearson correlation tests demonstrated significant positive associations between the entertainment subscale and two key engagement metrics: total gameplay hours ($r = 0.572$, $p = .001$) and number of quiz sessions ($r = 0.598$, $p < .001$).

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Table 4. MEEGA+ scores across all categories (maximum score: 4)

Variables	Mean	SD	Maximum	Minimum
Total usability	3.43	0.25	4.00	2.90
Aesthetics	3.43	0.54	4.00	1.50
Learnability	3.56	0.42	4.00	2.33
Functionality	3.51	0.45	4.00	2.50
Accessibility	3.29	0.49	4.00	1.67
Total player experience	2.95	0.49	4.00	1.81
User error protection	2.97	0.89	4.00	0
Confidence	3.25	0.60	4.00	1.00
Challenge	3.03	0.62	4.00	1.33
Satisfaction	2.85	0.69	4.00	1.25
Player social interaction	2.92	0.68	4.00	1.00
Entertainment	2.89	1.16	4.00	0
Focused attention	3.03	0.58	4.00	1.67
Communication	3.07	0.71	4.00	1.25
Perceived learning	2.91	0.76	4.00	1.33

Qualitative data were collected through open-ended responses from the MEEGA+ questionnaire, completed by 60 undergraduate students from Health Information Technology and Speech Therapy programs. Thematic analysis was conducted using Braun and Clarke's six-phase framework, which enabled the identification of dominant patterns and meaningful subthemes within the data.

The analysis revealed five major themes, each capturing frequently reported perceptions and suggestions related to the MedQuiz serious game. Together, these themes provide a comprehensive account of students' experiences, encompassing pedagogical benefits, usability considerations, and recommendations for improvement.

Theme 1: Competitive and Motivational Design

A significant number of participants (n = 25) highlighted the motivational impact of competitive features such as leaderboards, scoring systems, and multiplayer functionality. These elements were perceived as enhancing engagement and maintaining focus:

"The competition part makes it feel like a game, not just a quiz — it keeps me focused." "I liked seeing my rank and trying to beat my previous score."

Theme 2: Terminology Learning and Educational Impact

A recurrent theme (n = 20) emphasized the game's effectiveness in supporting medical terminology acquisition. Students valued the repeated exposure to key terms, the short-session learning format, and the ability to retain content more easily:

"It helps me learn faster — I repeat the terms without even realizing it." "Even when I make mistakes, I remember them better later."

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Theme 3: Interface and Visual Appeal

Approximately one-third of students (n = 18) praised the game's visual design, noting its appealing color schemes, smooth animations, and overall aesthetic simplicity:

"The design is smooth and attractive, not boring like many educational apps." "High-quality graphics helped me enjoy studying."

Theme 4: Desire for Personalization and Flexibility

Several participants (n = 15) expressed interest in greater customization, including adjustable difficulty levels, offline access, solo gameplay options, and personalized themes:

"I'd like to play at different levels depending on my knowledge." "An offline mode would help when internet is unstable."

Theme 5: Request for Enhanced Feedback and Scaffolding

A considerable number of students (n = 21) requested improvements in feedback mechanisms, particularly the ability to view correct answers immediately after incorrect responses and to review missed questions afterward:

"It would be helpful to show the correct answer after each question." "Let me review all the questions I got wrong — maybe in a summary page."

Importantly, no overtly negative sentiments were identified. Approximately 30% of responses were neutral or minimal in content (e.g., "no suggestions" or "no") and were therefore excluded from thematic categorization.

During the study, a technical vulnerability was identified whereby users could exploit the system by running the serious game concurrently on two devices—such as a smartphone and a laptop—to gain an unfair advantage in answering questions. In response, corrective actions were promptly implemented. Simultaneous login across multiple devices was disabled, and the database was updated to ensure real-time recording of responses following each question. These measures were designed to preserve the integrity of the game's competitive structure and to promote fairness throughout the learning process.

Discussion

This study evaluated the impact of a digital serious game, MedQuiz, on medical terminology acquisition and user satisfaction among healthcare students. Students who used MedQuiz demonstrated meaningful improvements in terminology knowledge, indicating that digital serious games can offer effective learning alternatives in medical education. High usability and engagement ratings from expert and student evaluations suggest that the game design resonated well with end-users, offering an accessible and motivating experience. These findings affirm the educational value of MedQuiz as a well-structured, evidence-informed digital serious game.

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Compared to other digital serious games in medical education, MedQuiz offers several unique advantages in platform compatibility, interaction design, and evaluation rigor. Unlike TERMInator [3]—a Moodle-based tool relying on static progress bars and sequential content unlocking—MedQuiz is a cross-platform system (web, Android, and PWA) that features real-time multiplayer functionality via WebSocket, a centralized instructor dashboard, and a modular structure ready for AI-based personalization. While TERMInator emphasizes ease of use, it lacks features such as individualized performance tracking and formal outcome evaluation. In contrast, MedQuiz demonstrated significant learning gains ($P < 0.001$) through a robust RCT, along with adaptive feedback and personalized progress analytics. In contrast to HistoRM [25]—a Unity-based serious game focused on oral histology within SCORM-restricted Moodle environments—MedQuiz applies a dynamic vocabulary reinforcement model rooted in clinical scenarios. Whereas HistoRM offers isolated game sessions with minimal peer interaction, MedQuiz promotes sustained engagement through its competitive multiplayer ecosystem, including weekly leaderboards and adaptive gameplay. The extended 8-week intervention period in MedQuiz, coupled with its interactive design, likely contributed to deeper retention and motivation, as evidenced by high SUS (90.36%) and HEP engagement scores (4.83/5). A key methodological distinction of this study lies in its use of a RCT, which contrasts with the less rigorous designs of previous digital serious game studies. For instance, TERMInator [3] relied on descriptive surveys without statistical evaluation of learning gains, while HistoRM [25] adopted a short-term crossover model without long-term follow-up. In contrast, *MedQuiz* applied a robust statistical framework including ANCOVA and multivariate regression, alongside validated instruments such as SUS, MEEGA+, and HEP to comprehensively assess learning impact and UX. Furthermore, as previously discussed, the game's design was informed by Flow Theory and SDT, which shaped both its motivational mechanics and user-centric structure. Unlike the digital board game examined by Tan et al. [23], which failed to show significant delayed learning outcomes despite high Flow scores, *MedQuiz* emphasized spaced repetition and sustained user engagement. Participants engaged with the game weekly for eight weeks, averaging 3.5 hours and seven competitive rounds per week—features that supported continuous reinforcement of terminology knowledge. While delayed post-tests were not included in this trial, the consistent engagement observed suggests preliminary evidence of learning consolidation. This design aligns with Flow Theory's emphasis on sustained immersion and SDT's emphasis on autonomous motivation. To strengthen long-term retention in future iterations, integrating AI-driven spaced repetition and post-intervention follow-up assessments is recommended. To elaborate further, Flow Theory, as articulated by Csikszentmihalyi, emphasizes the importance of maintaining an optimal balance between challenge and skill to foster deep engagement and intrinsic motivation. In MedQuiz, this was operationalized through features such as time-pressured rounds, and immediate feedback. The inclusion of a competitive multiplayer format with dynamic leaderboards also contributed to the "autotelic" experience described in Flow Theory, making gameplay enjoyable for its own sake while maintaining cognitive focus [26, 27].

Similarly, SDT, which highlights the three core psychological needs of autonomy, competence, and relatedness, directly informed several design choices [28, 29] . The game allowed autonomous navigation through topics, empowering learners to self-select quiz categories and difficulty levels. Competence was supported by real-time performance feedback, progress tracking, and incremental challenge scaling—designed to foster a sense of mastery. To promote relatedness, the game enabled social interaction via peer-vs-peer gameplay, visible ranking systems, and achievement sharing—all of which contributed to a socially meaningful and motivational learning environment.

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These theoretical foundations were not only integrated during the technical design phase but also used to interpret both quantitative (e.g., SUS, MEEGA+ scores) and qualitative feedback (e.g., student narratives emphasizing engagement, self-confidence, and competition). By anchoring MedQuiz in these well-established psychological theories, we ensured that the game's motivational architecture aligns with evidence-based principles of learning and user engagement.

The marked post-test improvement in the intervention group (28.23 ± 8.74) relative to the control group (20.8 ± 7.79, $P < .001$) provides compelling evidence for the educational effectiveness of digital serious games like *MedQuiz*. These results align with existing literature showing that serious games enhance knowledge acquisition, engagement, and retention in medical education. For example, a meta-analysis by Zhao et al. [30] confirmed that gamified platforms integrating feedback and adaptivity outperform traditional instruction. Likewise, Seidlein et al. [3] highlighted the value of interactive tools in terminology learning.

These findings are consistent with prior evidence demonstrating the pedagogical benefits of serious games in health professions education. For instance, Demir et al. [31] reported that students using a mobile-based anatomy game achieved significantly higher post-test scores while experiencing lower cognitive load. Similarly, Tan et al. [23] observed greater engagement and knowledge retention among students using an interactive game compared to a traditional e-learning module. In the context of advanced life support (ALS), Phungoen et al. [10] found that game-based instruction resulted in superior retention and learner satisfaction. Likewise, Felszeghy et al. [15] highlighted improved performance and motivation among histology students using gamified learning platforms. Collectively, these studies reinforce the view that well-designed serious games can enhance both cognitive and affective learning outcomes in medical training.

While a growing body of literature supports the pedagogical value of serious games in medical education, some studies have reported inconsistent results [9-14]. For instance, Tan et al. [23] observed no statistically significant gains in knowledge recall following the use of a gamified anatomy platform, even though students reported increased engagement and motivation. Such discrepancies may be attributed to variations in study duration, feedback mechanisms, or the degree of interactivity embedded in the game design. In contrast, MedQuiz featured real-time performance analytics and tailored feedback loops, which likely contributed to the superior learning gains observed in our trial.

Unlike Tan et al. [23], who found reduced retention in delayed post-tests conducted four weeks after a one-time intervention ($P = 0.75$), the present study involved consistent weekly gameplay sessions—averaging 3.53 hours and seven competitive rounds per week over an 8-week period—thereby fostering more sustained engagement. Although delayed assessments were not included in this trial, such extended interaction likely supported early phases of long-term knowledge consolidation. Furthermore, pattern analysis of engagement behavior provided additional insight into the learning mechanism. Students who consistently interacted with MedQuiz across all 8 weeks demonstrated significantly greater post-test gains compared to those with sporadic usage. This finding aligns with Schmitt's incremental learning theory and reinforces the pedagogical importance of repeated exposure and distributed practice in vocabulary retention [29]. While this evidence is preliminary, it suggests that sustained gameplay may facilitate deeper learning through consolidation and retrieval practice—both essential in mastering medical terminology. Future

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studies incorporating delayed post-tests are recommended to validate this consolidation effect over time. In addition to this group-wise comparison, correlation analysis further underscored the relationship between engagement intensity and learning outcomes. Both the number of games played ($r = .726$, $p < .001$) and the number of hours spent using MedQuiz ($r = .715$, $p < .001$) were strongly associated with higher post-test scores. These large effect sizes suggest that repeated exposure and greater time-on-task were significant contributors to knowledge acquisition. The strong positive correlation between these engagement metrics and performance not only supports the game's instructional design but also reinforces cognitive theories emphasizing distributed practice and retrieval-based learning. These findings complement prior studies highlighting how usage intensity in digital learning environments predicts deeper conceptual mastery and skill retention [9, 17-19]. In addition to time-on-task, learner motivation played a critical role, as evidenced by both quantitative and qualitative measures. The MEEGA+ instrument revealed strong scores in "Confidence" ($M = 3.25$), "Focused Attention" ($M = 3.03$), and "Player Social Interaction" ($M = 2.92$), pointing to the motivational influence of multiplayer design and social dynamics. Although the entertainment subscale of the MEEGA+ received a comparatively lower mean score (2.89), its relevance to engagement was nonetheless substantiated through correlation analysis. Pearson coefficients revealed moderate yet statistically significant associations between perceived enjoyment and both gameplay duration ($r = 0.572$, $p = .001$) and number of sessions played ($r = 0.598$, $p < .001$). These results indicate that students who experienced higher levels of fun were more likely to engage consistently with the platform. While entertainment was not the top-rated dimension of UX, its motivational influence appears to have supported continued participation—reinforcing the role of affective engagement in learning persistence. Future iterations of MedQuiz may benefit from integrating enhanced gamification elements to further amplify enjoyment without compromising cognitive focus. Beyond quantitative motivation scores, our thematic analysis revealed nuanced student perceptions that align with and extend Self-Determination Theory. Competitive features were widely cited as motivational, yet students also emphasized the importance of design aesthetics, error-based learning, and flexibility in gameplay. These insights support recent qualitative findings by Abou Hashish et al. [32], who identified motivation, peer learning, and visual appeal as primary drivers of engagement in game-based learning (GBL) environments across health sciences. Notably, students in our study requested greater personalization—such as difficulty adjustment and offline access—highlighting an emerging demand for learner-controlled pacing. This resonates with Van Gaalen et al. [33], who cautioned that GBL's effectiveness is contingent on perceived relevance and autonomy. While enjoyment is central to gameplay, students also expressed the need for justifying "play" in learning contexts, particularly in demanding or abstract courses like medical terminology. These findings collectively suggest that MedQuiz's educational impact lies not only in its competitive mechanics but also in its capacity to provide meaningful, efficient, and flexible learning experiences. Future design iterations should therefore incorporate optionality, adaptive feedback, and context-sensitive scaffolding to balance cognitive load, enhance perceived utility, and sustain engagement across diverse learner profiles.

Multivariate regression analysis provided additional insights into predictors of post-test performance within the digital serious game environment. Students who engaged more deeply with MedQuiz tended to benefit the most, emphasizing the role of sustained motivation and personalized interaction in educational outcomes ($B = 8.16$, $P = .006$), explaining 74.3% of the

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variance (Adjusted R^2 = .645). This underscores the critical role of intrinsic motivation and user-centered gameplay design in driving educational outcomes in digital serious games.

A significant difference was observed across academic disciplines, with HIT students outperforming Speech Therapy peers in post-test scores (B = -10.066, p = .004), despite similar baseline performance. This disparity likely stems from HIT curricula offering structured exposure to digital health tools, artificial intelligence, and app-based learning, fostering greater technological fluency and receptiveness to gamified platforms. These interpretations are consistent with competency frameworks emphasizing digital literacy in health education, such as those from the Health and Care Professions Council [34] and the QAA Benchmark Statement for Biomedical Sciences [35], which highlight the importance of data fluency in modern healthcare training [36]. Supporting this view, Baigi et al. [37] reported that HIT students interacted more effectively with digital nutrition apps than peers from other fields, and Mousavi Baigi et al. [38] similarly found that paramedical students—including those in HIT—demonstrated stronger conceptual familiarity with AI than clinical counterparts. Broader research further shows that prior experience with digital tools enhances engagement with serious games [39, 40]. While MedQuiz provided tutorials, videos, and technical support, learners from less tech-oriented backgrounds may benefit from additional scaffolding—such as adaptive onboarding, microlearning modules, or visual cues—to ensure equitable engagement across disciplines.By contrast, variables such as pre-test scores, GPA, gender, perceived usability, and frequency of gameplay (digital or otherwise) did not significantly predict learning outcomes. The non-significance of gender is particularly notable. Subgroup analyses likewise revealed no gender-based differences in gameplay duration or participation frequency, nor in performance metrics. This reinforces the inclusive and motivational structure of MedQuiz—grounded in Self-Determination Theory and Flow Theory—which emphasized autonomy, real-time feedback, and social interaction. These core features appear to have effectively bridged traditional gender gaps in digital learning environments [1, 41].

Comparative analysis with other digital serious games further highlights MedQuiz's pedagogical strengths. For instance, unlike Tan et al. [23], whose gamified anatomy platform showed no significant improvement in knowledge retention (P = 0.75), MedQuiz achieved sustained learner engagement over an 8-week period, with participants averaging 3.5 hours of gameplay per week and completing seven competitive rounds. This level of longitudinal interaction suggests a greater potential for durable knowledge consolidation—underscoring the critical role of repetition and ongoing engagement in effective serious game–based learning.

In addition, findings from the MARS-based evaluation conducted by Mousavi Baigi et al. [42] reveal that MedQuiz outperforms many reviewed applications in areas such as performance, aesthetics, and interactivity. Unlike most apps in that study—which lacked dynamic feedback mechanisms and analytical dashboards—MedQuiz integrates real-time performance tracking, immediate feedback, and synchronous multiplayer features enabled by WebSocket technology. These functionalities distinguish MedQuiz from other digital serious games such as TERMInator, HistoRM, and MediTerm. Finally, the systematic review by Al Habsi et al. [43] on serious games in anatomy education underscores the cross-disciplinary potential of such tools. Key domains such as knowledge acquisition, learner engagement, and skill development were consistently improved through digital serious games—dimensions that closely align with MedQuiz's architecture and pedagogical goals. These findings support the feasibility of adapting MedQuiz beyond medical

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terminology to areas like anatomy, pharmacology, and pathology by incorporating modular, scenario-based vocabularies tailored to each discipline.

This study provides valuable insights into the role of digital serious games in medical education and represents one of the first RCTs specifically evaluating a digital serious game developed for medical terminology acquisition among healthcare students. Unlike previous studies that predominantly focused on traditional e-learning methods, MedQuiz was systematically assessed through a combination of expert usability evaluations and real-world user feedback, ensuring a comprehensive and balanced analysis of its educational impact.

One of MedQuiz's key strengths lies in its modular and adaptable architecture, which facilitates expansion beyond medical terminology to include other disciplines such as anatomy, pharmacology, and pathology. This flexibility offers a significant advantage over conventional learning resources that often lack interactivity and personalization. Moreover, MedQuiz's cross-platform accessibility (web-based and mobile) enables learners from diverse socioeconomic backgrounds to benefit from its features without the need for specialized hardware or software.

The study design further strengthens its findings. Through rigorous usability testing using the SUS and HEP frameworks, MedQuiz demonstrated high levels of engagement, usability, and motivational impact. Features such as real-time feedback and a dynamic leaderboard system contributed to a highly interactive and immersive learning environment, reinforcing terminology retention and user satisfaction.

Nonetheless, several limitations must be acknowledged. First, the participant pool was restricted to students in HIT and Speech Therapy programs, which may limit the generalizability of the findings to other disciplines such as nursing, medicine, or pharmacy. Additionally, the significant differential effect by academic discipline—whereby HIT students outperformed their Speech Therapy peers despite equivalent baseline performance—suggests that the intervention may be more readily adopted or effective in fields with stronger digital readiness. Therefore, the results from this study are most applicable to Health Information Technology students; transferability to other healthcare disciplines requires further validation through targeted studies. Second, the current study specifically assessed immediate post-intervention learning outcomes. No delayed post-tests were conducted; therefore, conclusions regarding long-term knowledge retention cannot be drawn. Although students maintained weekly engagement during the 8-week intervention, future trials should include follow-up assessments at 4–8 weeks post-intervention to evaluate the durability of learning gains over time. Third, since the study was conducted at a single academic institution, there remains a possibility of information exchange between the intervention and control groups (contamination bias). To mitigate this, single-use installation codes were implemented; however, future multi-site trials are recommended to enhance external validity and reduce potential data contamination.

Fourth, while usability scores were high, qualitative feedback from students indicated areas for improvement—particularly regarding user engagement and gameplay variation. Suggestions included the integration of multiplayer tournaments, discussion forums, and adjustable difficulty settings. Incorporating AI-driven personalized learning pathways could also improve learner adaptability and motivation. Furthermore, although student feedback was collected anonymously and without any academic grading implications, we acknowledge that the serious game was introduced by an instructor affiliated with the study. This instructional role may have introduced social desirability bias in participants' evaluations. Future studies should consider involving third-

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party facilitators to administer the intervention and its evaluation, thereby minimizing potential bias related to instructor influence.
Also, although MedQuiz was used alongside traditional lectures, the study did not directly compare its effectiveness with other contemporary digital learning platforms such as adaptive e-learning systems, AI-powered tutors, or augmented reality (AR) applications. Future comparative effectiveness trials are necessary to position MedQuiz within the broader ecosystem of digital health education tools.
Finally, although the sample size ($n = 60$) was adequate to detect statistically and practically significant effects—including large effect sizes for both primary (Cohen's $d = 0.90$) and secondary outcomes ($f^2 = 2.89$)—the number of predictors in the regression model ($k = 8$) relative to the intervention group sample ($n = 30$) raises the possibility of overfitting. To address this, a split-sample cross-validation was conducted, which confirmed the model's stability (R^2 reduced moderately from .743 to .599). Nonetheless, replication with larger samples is recommended to further enhance model generalizability.

To enhance its scalability and educational value, several development strategies are proposed for MedQuiz. Its modular structure already supports the integration of additional clinical domains—such as anatomy, pharmacology, and pathology—through dynamic content management and customizable vocabulary scenarios. This design allows instructors to tailor game content based on specific curricular needs, expanding its utility across disciplines.
Incorporating artificial intelligence (AI) into the platform would enable adaptive learning, allowing MedQuiz to respond to individual performance metrics and guide learners through personalized content pathways. Enhanced game mechanics, including real-time team-based challenges, leaderboard incentives, and achievement systems, could further increase user motivation and sustained participation.
From a systems integration standpoint, MedQuiz supports interoperability with leading learning management systems (LMS), such as Moodle and Blackboard, via SCORM compatibility and API endpoints. This makes the platform suitable for blended and flipped classroom models. Additionally, the development of multilingual versions would facilitate adoption across diverse educational settings and geographic regions.
Future research should explore not only the platform's ability to improve knowledge acquisition but also its effectiveness in promoting clinical decision-making and practical skills. Longitudinal studies assessing delayed retention, as well as implementation in interdisciplinary curricula, will be crucial in validating MedQuiz's broader educational potential. Furthermore, multi-site RCTs across institutions and healthcare disciplines (e.g., nursing, medicine, rehabilitation) are recommended to enhance external validity and assess generalizability across educational contexts. Given its modular architecture and customizable content framework, MedQuiz can be readily adapted to align with terminology and curricular demands in various allied health programs.

Overall, this study highlights the transformative potential of MedQuiz as a scalable, evidence-based learning tool that supports curricular goals, promotes equitable access, and enhances terminology acquisition across health education contexts. Future developments should focus on long-term retention, integration with adaptive learning strategies, and broader implementation across disciplines.

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Methods

Study Design

This study was designed as a double-blind, parallel-group, RCT, conducted in accordance with the CONSORT. The primary objective was to evaluate the effectiveness of a digital serious game, MedQuiz, in enhancing medical terminology learning among health sciences students. Participants were randomly assigned to one of two groups: the intervention group, which engaged with the MedQuiz, digital serious game alongside traditional instruction, and the control group, which received traditional lecture-based instruction only. The complete RCT protocol, outlining the full methodological framework of this study, is available for reference [24]. The duration of the intervention was determined based on the academic structure of the Medical Terminology course, which spans approximately four months in a standard semester. To ensure curricular alignment and educational equity, the intervention was implemented during the first half of the semester. This two-month period allowed for adequate exposure to the MedQuiz and timely post-intervention assessment. It also enabled researchers to provide the control group with access to the game during the second half of the term, ensuring equal learning opportunities for all participants before final examinations. Although all participants were eventually granted access to the serious game after the study, no within-subject comparison was conducted, as the control group did not undergo additional post-testing following their delayed access. Given the risk of carryover effects in educational interventions, a within-subject or crossover design was deemed inappropriate.

Participants and Randomization

The study population comprised undergraduate students enrolled in the Medical Terminology course at the School of Paramedical Sciences, MUMS. As this course was exclusively part of the curricula for HIT and Speech Therapy programs, only students from these disciplines were eligible to participate. Given the absence of prior research examining the efficacy of digital serious games for medical terminology learning, a convenience sampling approach was employed.

To ensure balanced group sizes and reduce selection bias, block randomization with a fixed block size of 4 was employed. The randomization sequence was generated prior to participant recruitment using SealedEnvelope.com, a secure online platform for computer-generated random allocation. The sequence was prepared by an independent coordinator who was not involved in the recruitment, enrollment, or assignment of participants.

Eligible students were randomly assigned to either the intervention group (MedQuiz) or the control group (traditional instruction) using a 1:1 ratio. Allocation concealment was maintained throughout the recruitment process by restricting access to the allocation list until after participant enrollment. This approach prevented both participants and study investigators from predicting group assignment, ensuring methodological rigor and minimizing selection bias.

The required sample size was calculated using the formula outlined by Steven A. Julious in *Sample Sizes for Clinical Trials* [44], as shown in equation (1):

$$(1)\ n = 2 * \frac{\left(\left(Z_{1-\beta} + Z_{1-\frac{\alpha}{2}}\right)^2 * \sigma^2\right)}{d^2}$$

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Where:

- α = 0.05 (95% confidence interval)
- β = 0.20 (power = 80%)
- σ = 2 (estimated standard deviation)
- d = 2 (minimum expected difference)

Substituting the values into equation (1):

$$n = 2 * \frac{((1.64 + 1.96)^2 * 2^2)}{2^2}$$

Accounting for a 10% potential dropout rate, the final sample size was adjusted to 30 participants per group, resulting in a total of 60 participants.
This estimation was informed not only by Julious's recommended formula [44] but also by standard deviation and effect size values reported in similar previous studies on digital serious games in medical education contexts [23, 25, 45].

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Eligibility Criteria
Participants in this study included undergraduate students in their second semester of the HIT program and fifth semester of the Speech Therapy program, actively enrolled in the Medical Terminology course at MUMS during the study period. The inclusion of these specific cohorts ensured curricular consistency and minimized confounding variables.
Students were excluded if they were transfer or guest students from other universities, as their prior exposure to medical terminology might have varied significantly. Additionally, those who declined to participate or lacked access to a smartphone capable of running the MedQuiz application were also excluded, as the intervention required digital engagement.

Blinding (Allocation Concealment)
Due to the nature of the intervention, participant blinding was not feasible. However, blinding procedures were implemented for researchers and data analysts to mitigate bias. The statistician performing the analyses was unaware of group assignments. Moreover, all assessments were coded, preventing evaluators from associating test results with specific groups.
This structured approach ensured methodological rigor while addressing the practical constraints of digital learning interventions.

Outcome Measures
The primary outcome of this study was the improvement in medical terminology knowledge, measured by the change in scores on a standardized multiple-choice Medical Terminology Assessment administered pre- and post-intervention. This assessment, originally developed as a 20-item test in the published protocol [24], was expanded to 40 expert-validated items prior to implementation to ensure broader coverage of medical terminology domains. The same 40-item version was used consistently across both groups.

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The secondary outcomes included: User engagement, assessed via automatically recorded gameplay metrics (e.g., time spent on the game, number of competition sessions, accuracy rates).
Learner experience, including usability, motivation, and perceived learning, was measured using the MEEGA+ questionnaire. System usability and playability were evaluated before the intervention through expert assessments using the SUS and the HEP frameworks.
These outcomes were selected to provide both objective and subjective insights into the educational effectiveness and user acceptance of the MedQuiz platform.

Intervention: Design and Development of MedQuiz Digital Serious Game
MedQuiz was designed as a multi-platform digital serious game aimed at improving medical terminology acquisition through digital serious games and interactive competition. Recognizing the increasing role of digital serious games in medical education, the game was developed to bridge existing educational gaps and enhance student engagement. The development process followed a structured four-phase model: needs assessment, content development, technical implementation, and evaluation.
The game was made accessible via both an Android mobile application and a web-based platform, ensuring seamless compatibility across iOS, laptops, and desktops. Additionally, a web-based administrative dashboard was developed to allow educators to monitor student activity, track learning progress, and manage user access in real-time. This ensured that the game was not only a learning tool for students but also an evaluation and management system for educators.

MedQuiz was designed with two primary components, allowing students to engage in both independent learning and interactive competition:
In the Self-Paced Learning Module, structured educational content was presented in the form of videos, PDF documents, and PowerPoint presentations. The materials were systematically categorized based on body systems, ensuring alignment with standard medical curricula. Unlike traditional educational platforms, this module was not built around sequential multiple-choice testing but instead allowed students to explore and review content at their own pace, encouraging personalized learning approaches.
The Multiplayer Competitive Mode fostered real-time, two-player competitive matches, reinforcing learning through active engagement and recall-based challenges. Each match consisted of six rounds, with three multiple-choice questions per round. Players took turns selecting question categories (e.g., cardiovascular system, gastrointestinal system), promoting both strategic decision-making and deeper retention of medical terminology. Progress tracking was embedded within the platform, with scores, performance metrics, and competitive rankings automatically recorded in individual player profiles.

1. By integrating these two components, MedQuiz provided a balanced learning experience, where students first acquired foundational knowledge independently and then reinforced their learning through structured competitive gameplay. This combination fostered greater motivation, engagement, and retention compared to conventional learning methods. **Figure 3 and Figure 4** illustrates the main interface of the game's administrative dashboard, where game management and user interactions can be monitored.
2. The development of the MedQuiz serious game followed a structured, multi-phase process that included design, technical implementation, and iterative testing. Several coordination meetings were conducted with the development team to determine key elements such as programming

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frameworks, game scenarios, platform selection (web vs. app), and database architecture. The major development stages are outlined below:

3. Wireframe Design: The initial phase involved creating wireframes to outline the structure and layout of the UI. This step provided a foundational blueprint for both functionality and navigation.
4. UI Design: The UI was designed using *Figma*, with a focus on creating an engaging and intuitive experience. The primary aim was to ensure that learners could easily navigate the digital serious game and interact seamlessly with its core features.
5. Frontend Development: The frontend was implemented using HTML5, CSS3, Tailwind CSS, and JavaScript frameworks such as Vue.js and Nuxt.js. This allowed for a responsive and visually appealing interface compatible with both web and mobile environments.
6. Database Design: PostgreSQL was used to develop a secure and optimized database capable of managing user profiles, performance metrics, and gameplay data efficiently.
7. Backend Development: Laravel 10 served as the backend framework, supporting secure communication between clients and servers, and handling critical functions such as game mechanics, data synchronization, and user management.
8. Admin Dashboard Development: A web-based administrative dashboard was developed to give instructors full control over content, gameplay sessions, and student progress. This system allowed for real-time monitoring, user account management, and question bank editing.
9. SMS Integration: An SMS-based messaging service was integrated into the platform to notify users of important updates and game-related alerts.
10. Real-Time Communication: WebSocket technology was implemented to enable real-time multiplayer interactions, ensuring immediate delivery of scores, gameplay feedback, and match results.
11. Progressive Web App (PWA) and APK Versions: MedQuiz was released as a PWA for web and iOS devices, and as an Android APK file. The mobile version was developed using React Native to ensure consistent performance across operating systems.
12. Beta Testing and Feedback: A beta version was released for a limited user group to identify technical issues, assess performance, and collect usability feedback prior to the full launch.
13. Quality Testing and Bug Fixes: Comprehensive testing was conducted following beta deployment, focusing on cross-platform functionality, UI consistency, and gameplay stability.
14. Final Release: The final stable version of the game was released, making it available for download via two methods: as an Android APK file (Download link) and as a web-based game (Web link). To further enhance UX, a video tutorial for the game was created, which can be accessed by scanning a QR code provided in the game interface (available from: https://www.aparat.com/v/lrlb350). **Figure 3 and Figure 4** provides an overview of the MedQuiz game interface, showing the main screen, player rankings, user profile, and ongoing match status.

In addition to the game itself, a web-based admin panel was developed to manage the game, monitor user activities, and generate reports (**Figure 5)**. This panel allowed instructors to manage user accounts, review performance, and make adjustments to the game content as needed (Access here).

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Access Control and Contamination Prevention

To minimize the risk of contamination between groups—a common concern in educational trials involving digital interventions—strict access control measures were implemented. Each participant in the intervention group was assigned a unique, one-time activation code that became invalid upon initial use. This mechanism ensured that students in the control group could not gain unauthorized access to the game during the study period, preserving the integrity of the experimental conditions. A centralized administrative dashboard was used to verify and approve access only for participants listed in the intervention group. Simultaneous login across platforms (e.g., web and app) was blocked to prevent parallel access or credential sharing. No registration was confirmed unless matched to the pre-approved user list based on student ID. Real-time logs of login and gameplay activity were monitored by the research team to ensure strict adherence to group allocation and prevent cross-contamination.

All game-related data—including time spent playing, number of sessions, and performance metrics—were automatically logged and securely stored in a centralized database. This enabled precise tracking of engagement levels and adherence to the intervention, allowing researchers to assess learning outcomes objectively.

Educational Impact and Theoretical Justification

The design of MedQuiz was informed by extensive research, direct user feedback, and a strong theoretical foundation to ensure alignment with student learning needs and best practices in digital serious games. In the Needs Assessment and Curriculum Alignment phase, a comprehensive review of medical curricula was conducted, ensuring that the game's content and structure were relevant to students in Health Information Technology and Speech Therapy programs. A survey of 496 students highlighted the need for interactive digital learning tools, showing that 75.7% of medical students and 66.7% of paramedical students used mobile devices for educational purposes [16], reinforcing the importance of mobile-friendly solutions tailored to students' technological preferences.

MedQuiz's design also draws on well-established learning theories to maximize pedagogical effectiveness. Grounded in constructivist principles, the game fosters active learning by promoting engagement rather than passive knowledge acquisition. Nation's [46] model of vocabulary knowledge, which emphasizes learning words through form, meaning, and use, guided the integration of medical terminology in an interactive format, where students engage in self-directed learning and competitive gameplay. Additionally, Schmitt [47] incremental learning theory was applied by progressively increasing quiz difficulty to enhance retention and reinforce knowledge acquisition. MedQuiz also incorporated cognitive theories such as Asher's [48]. Total Physical Response (TPR), which emphasizes physical interaction to enhance language acquisition. This informed the timed-response mechanics of the game, encouraging students to make quick lexical decisions, reflecting real-world medical scenarios. Furthermore, time constraints based on Chapelle's [49] research were integrated to develop both speed and accuracy in medical terminology retrieval.

Finally, the principle of automaticity in language learning [50] underpinned the game's goal of transitioning learners from effortful recognition to automatic retrieval, essential in fast-paced clinical environments. The inclusion of digital serious game elements, such as leaderboards and reward systems, was guided by Mueller [7], emphasizing motivation and sustained engagement. By combining these theoretical models, MedQuiz offers a scientifically structured, research-

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backed approach to medical terminology education that not only improves retention and engagement but also enhances clinical competence.

Pre-Intervention Validation of the MedQuiz Digital Serious Game

Before the implementation of the intervention, a comprehensive usability and playability evaluation of the MedQuiz platform was conducted by a panel of expert reviewers. The validation process aimed to assess the game's technical performance, UX, and educational effectiveness to ensure that it met the standards for serious games.

Usability Assessment

The SUS, originally developed by Brooke et al. [51], was employed to evaluate the ease of use, accessibility, and overall functionality of the MedQuiz platform. A panel of subject matter experts, including specialists in medical education, health informatics, instructional design, and game development, participated in the assessment. The SUS provides a standardized measure of usability, allowing for objective benchmarking of UX in digital applications.

Playability Evaluation

To assess the overall gameplay experience and engagement quality, the HEP framework, developed by Desurvire & Toth [52], was utilized. This framework evaluates critical elements such as game mechanics, feedback systems, user motivation, and the balance between challenge and reward. The expert panel conducted structured usability tests and provided feedback on various aspects of game mechanics, interface navigation, and learning efficacy.

Refinements and Optimization

Based on the expert feedback, several refinements were made to improve the UX, content delivery, and technical performance of the game. Adjustments included enhancing interface responsiveness, optimizing game mechanics for smoother real-time interaction, and refining the instructional content layout to further align with pedagogical best practices. These improvements ensured that MedQuiz provided a seamless, engaging, and pedagogically sound learning experience before its deployment in the RCT.

Implementation and Evaluation of Intervention Effectiveness

A two-month RCT was conducted to evaluate the effectiveness of the MedQuiz digital serious game in enhancing medical terminology learning. Undergraduate students enrolled in the HIT and Speech Therapy programs at MUMS, who had Medical Terminology as a core component of their curriculum, were randomly assigned to either the intervention or control group.

Both groups participated in traditional classroom-based instruction. However, the intervention group was additionally provided access to the MedQuiz game for the duration of the study, integrating digital serious games into their routine educational activities. All educational content embedded within the MedQuiz platform, including instructional materials and quiz questions, was identical to the resources available to the control group in the form of lecture notes and handouts. This ensured that the only difference between the groups was the digital serious game component, serving as an instructional supplement rather than a content replacement.

To prevent contamination between groups, each participant in the intervention arm received a unique, one-time activation code for the MedQuiz platform, which became inactive after initial use. This measure ensured that control group participants could not access the game during the intervention phase. However, to maintain educational equity, control group participants were

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granted access to the game after study completion, allowing them to benefit from the intervention post-trial.
Knowledge acquisition was assessed pre- and post-intervention using a standardized Medical Terminology Assessment, consisting of 40 multiple-choice questions validated by a panel of subject matter experts. This assessment also collected demographic data and student GPAs. The primary outcome measure was the difference in test scores between the pretest and posttest in both groups.
To evaluate adherence and engagement levels within the intervention group, automated data tracking was implemented. The system recorded game usage duration, frequency of gameplay, number of competitions participated in, and accuracy rates in answering quiz questions. These logs provided an objective measure of student engagement and compliance with the intervention protocol.

Following the intervention period, participants' motivation, UX, and learning engagement were assessed using the MEEGA+ questionnaire, a validated instrument developed by Petri & von Wangenheim [53]. This questionnaire measures multiple dimensions of digital serious game experiences, including motivation, perceived learning effectiveness, engagement, and usability.

Data Analysis and Statistical Reporting

The normality of the data distribution was assessed using both the Shapiro–Wilk test and visual inspection of histograms and Q-Q plots. Statistical analyses were conducted using SPSS version 27 with a 95% confidence interval (CI) and a two-tailed significance level of $p < 0.05$. Descriptive statistics, including means and standard deviations, were calculated and presented using tables and visualizations. For between-group comparisons, independent t-tests were applied for normally distributed continuous variables, while Mann–Whitney U tests were used for non-normally distributed data. Chi-square tests assessed categorical variables. To control for baseline differences, analysis of covariance (ANCOVA) was employed to isolate the intervention effect from potential confounders. Furthermore, a multiple linear regression analysis was conducted to explore the predictive effects of demographic and academic variables (e.g., pre-test scores, field of study, GPA, gender, gaming habits) on post-test performance. Additionally, responses from the questionnaire were analyzed qualitatively using thematic coding, allowing for an in-depth exploration of UX and perceptions.

Ethical Considerations

This study was approved by the Ethics Committee of Mashhad University of Medical Sciences (MUMS), under Approval ID: IR.MUMS.REC.1400.336, dated 29 January 2022. The complete RCT protocol, outlining the study's full methodological framework, is published and available for reference [24].

Before participation, all students received a detailed explanation of the study's objectives, procedures, and voluntary nature by a designated data collector. Written informed consent was obtained from each participant. Informed consent for publication was also obtained. No clinical photographs, videos, or personally identifiable health data were collected or published. The only participant-related element visible in the intervention was a self-selected in-game username, which

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could, in rare cases, be recognizable to peers. All participants were clearly informed of this possibility and gave explicit, voluntary consent for the display of their chosen usernames within the game environment.

Each participant was assigned a unique research identification code, which was linked to their personal information (e.g., name and contact details) but stored separately to ensure confidentiality. Identifiable data were securely stored on encrypted institutional servers in password-protected files, with access restricted to the ethics-approved research team. The data collected through surveys and the MedQuiz platform included only de-identified research IDs, ensuring anonymity during analysis.

No financial compensation was provided to participants. However, as an educational incentive, the intervention group was granted full access to the MedQuiz game throughout the semester. To maintain educational equity, students in the control group were also granted access to the game after the study concluded, allowing them to benefit equally in preparation for final exams.

All study procedures were conducted in accordance with the ethical standards of the institutional and/or national research committees and with the principles outlined in the Declaration of Helsinki (1964) and its subsequent amendments.

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Declarations

Data availability: Deidentified data that support the findings of this study are available on request from MS or SFMB.

Code availability: The source code of the MedQuiz platform is not publicly available, as it was developed as a proprietary application for internal research and educational purposes. However, researchers interested in the technical architecture or implementation details may contact the corresponding author for further information.

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Authors' Contributions: KK supervised the project, acquired funding, and coordinated the technical development of the MedQuiz game. SFMB administered the study, coordinated protocol development, and drafted the manuscript. MS contributed to the study design, supervised implementation, and helped revise the methodology. SFMB and MS contributed equally to this work. DS facilitated participant recruitment in the Speech Therapy group and contributed to assessment design. AD supported data collection, contributed to question development, and acted as a liaison with the game development team. PM and MKR conducted the statistical analysis and validated the data. SMT developed the software infrastructure and provided technical guidance. MSF designed and implemented

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the MedQuiz platform. ME coordinated participant data collection. KK, SFMB and MS also authored the RCT protocol. All authors reviewed and approved the final manuscript.

Competing Interests: The writers have stated that they have no competing interests.

Trial Registration: The study protocol has been published (DOI: [10.3233/SHTI220658]; published date: 2022, Jun 29).

Abbreviations

CONSORT: Consolidated Standards of Reporting Trials
GPA: Grade Point Averages
HEP: Heuristic Evaluation of Playability
HIT: Health Information Technology
MEEGA+: Metrics for Educational Game Assessment+
MUMS: Mashhad University of Medical Sciences
RCT: Randomized Controlled Trial
SUS: System Usability Scale
UI: User Interface
UX: User Experience
ALS: Advanced Life Support
SDT: Self-Determination Theory

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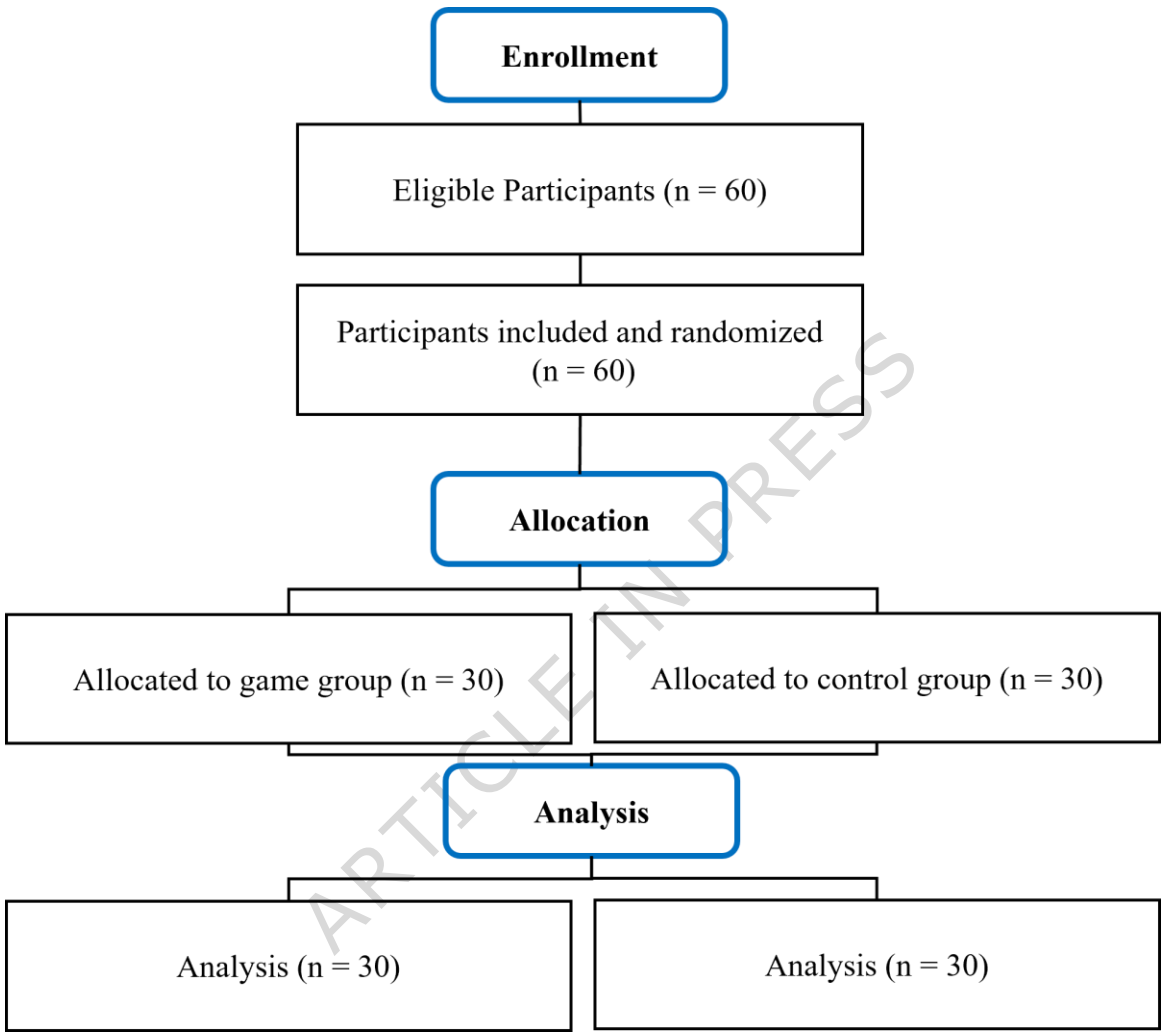
Figure Legend

Figure 1. Participant Flow Diagram



This figure illustrates the flow of participants throughout the study, from recruitment to allocation, intervention, and data analysis. It includes the number of students screened, randomized, and retained in each group.

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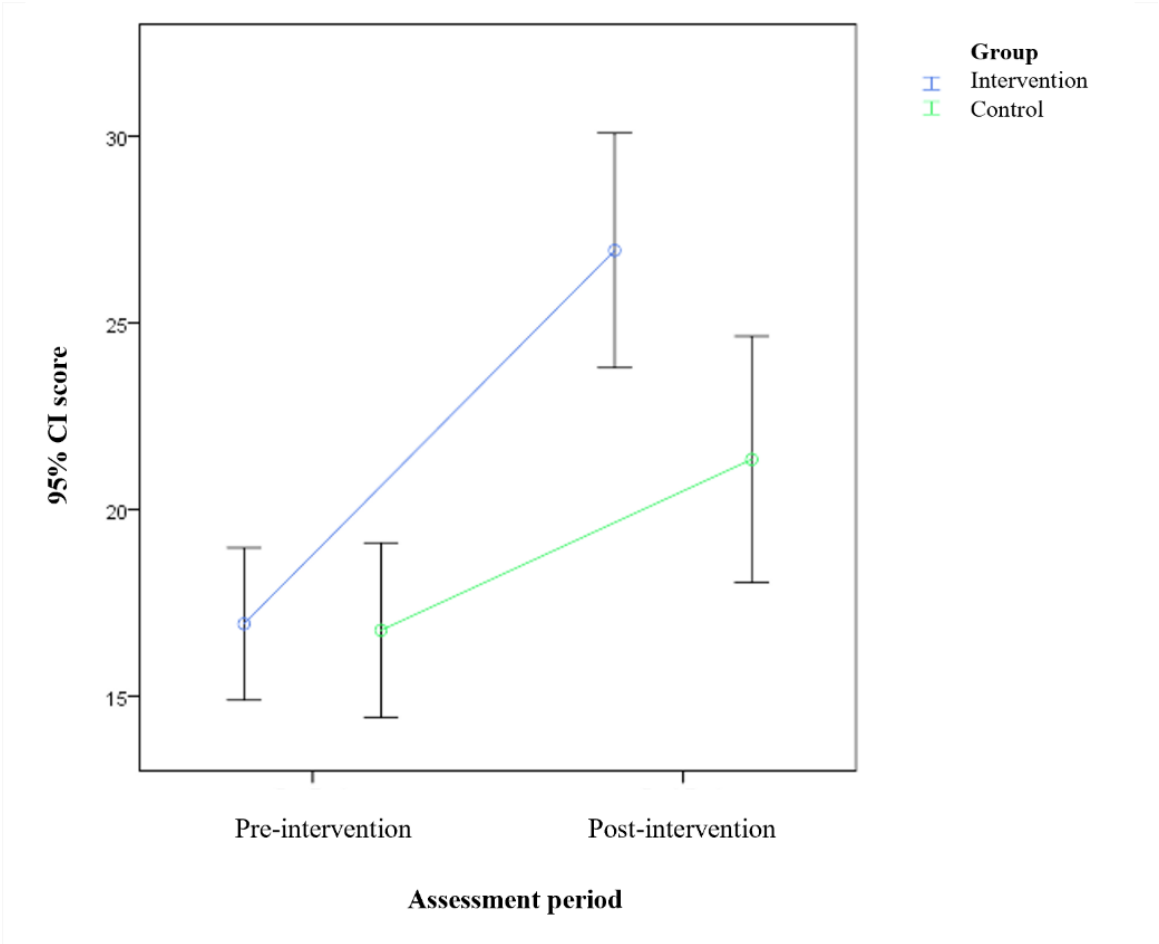
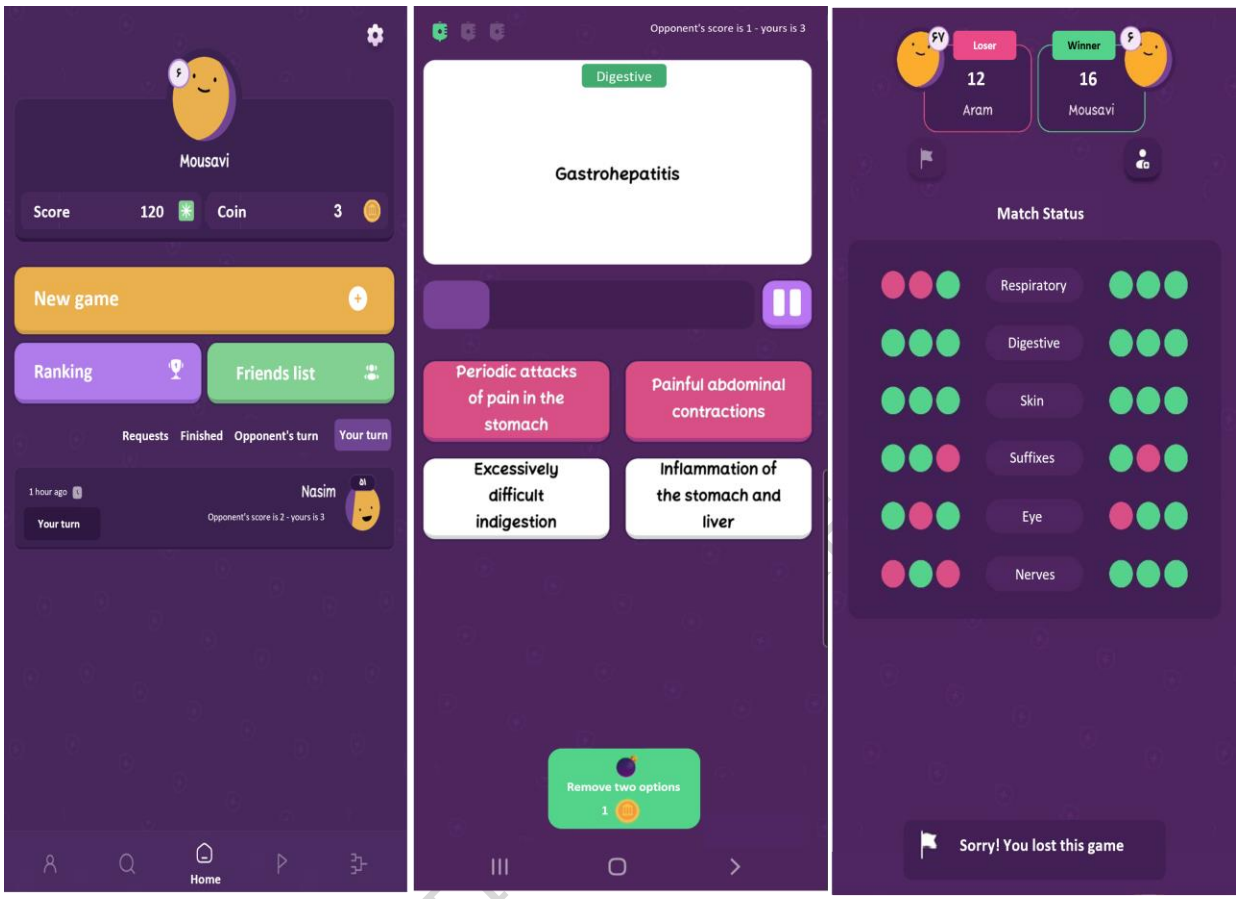
Figure 2. Changes in medical terminology scores before and after the intervention



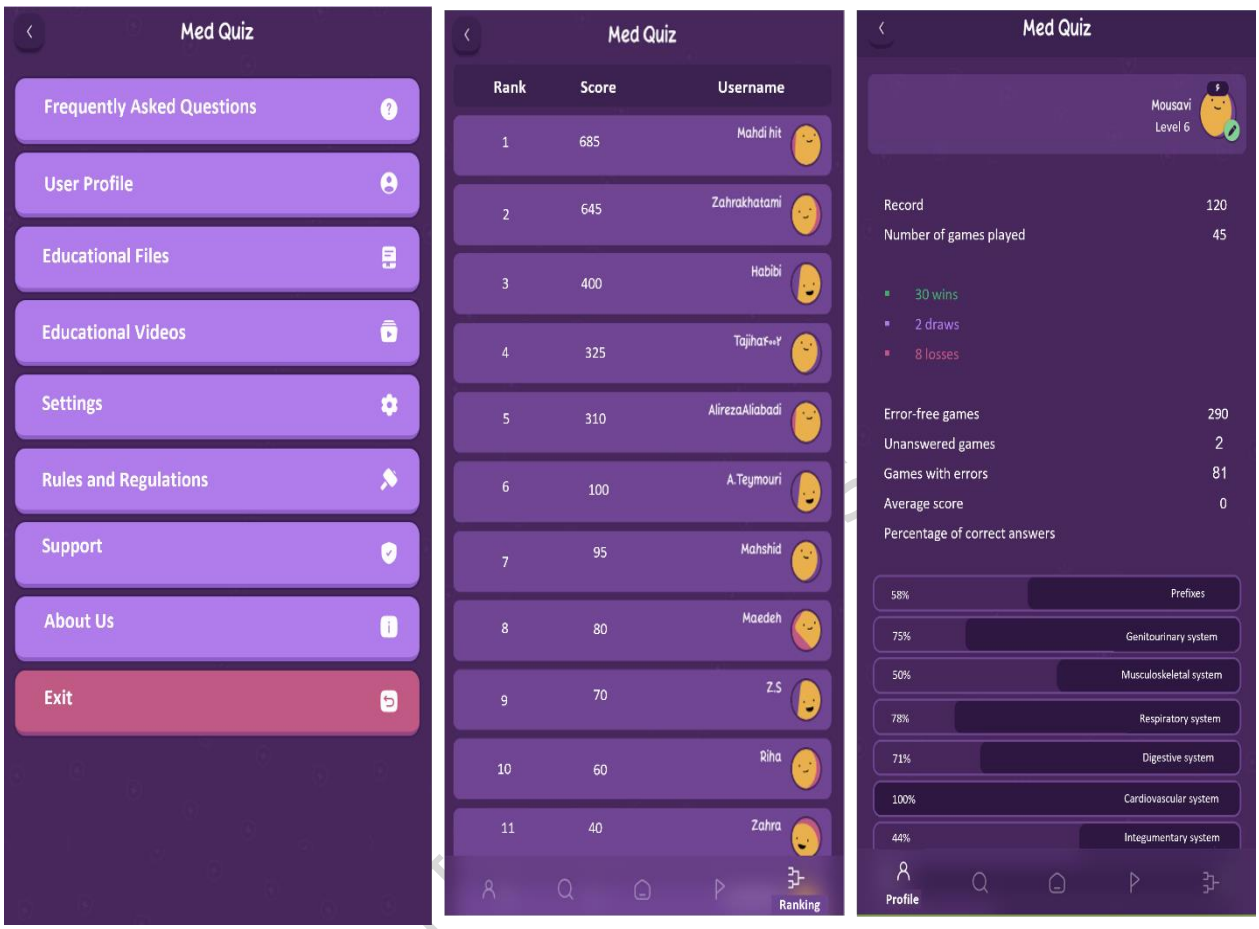
The bar chart compares pre-test and post-test scores in both groups. The intervention group showed a substantial improvement (17.13 ± 6.11 to 28.23 ± 8.74), while the control group had a smaller gain (16.60 ± 5.51 to 20.80 ± 7.79).

Figure 3. Screenshots of the MedQuiz digital serious game interface



Panel (a) displays the home screen with access to game modes, score tracking, and friend challenges. Panel (b) shows an in-game decision support feature that removes incorrect options using coins. Panel (c) presents a post-game match summary comparing correct and incorrect answers and final outcomes.

Figure 4. Key interactive features supporting personalization and motivation



Panel (a) shows the main game menu with categorized resources, videos, and user support options. Panel (b) illustrates the leaderboard that tracks rankings and promotes competitive engagement. Panel (c) presents a user profile dashboard where players monitor progress and select targeted practice modules.

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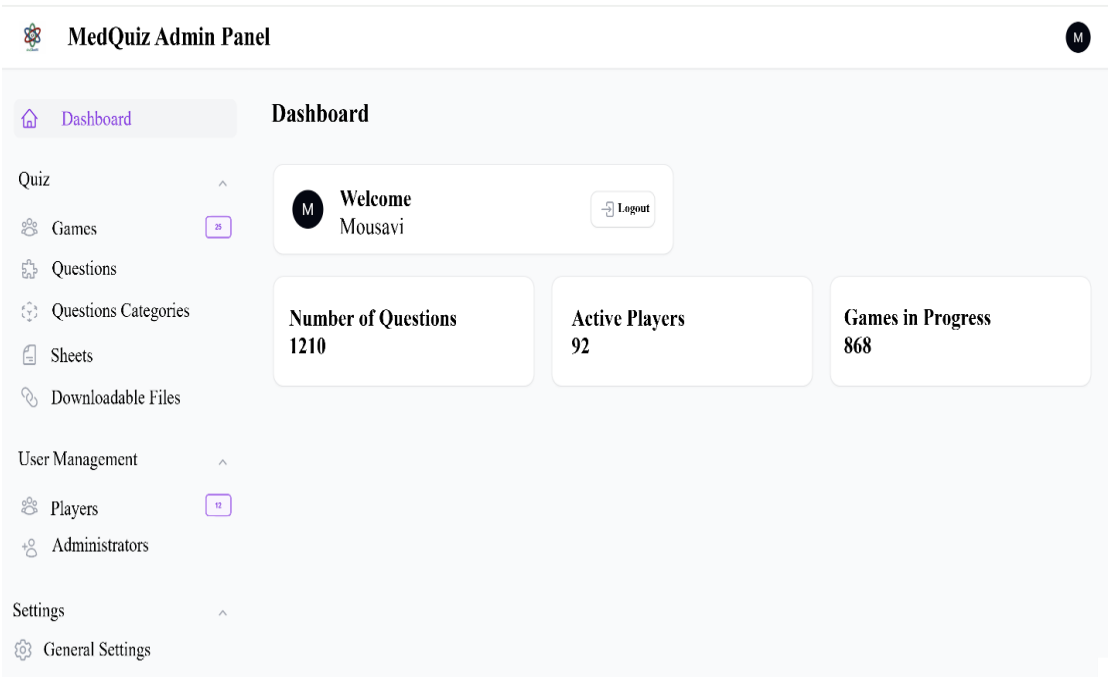
Figure 5. The MedQuiz administrative dashboard for educators



This dashboard enables instructors to manage game content, categorize questions, and upload multimedia resources. It includes reporting tools for tracking game progress and learner performance, and features for assigning administrative roles and organizing content by body systems.

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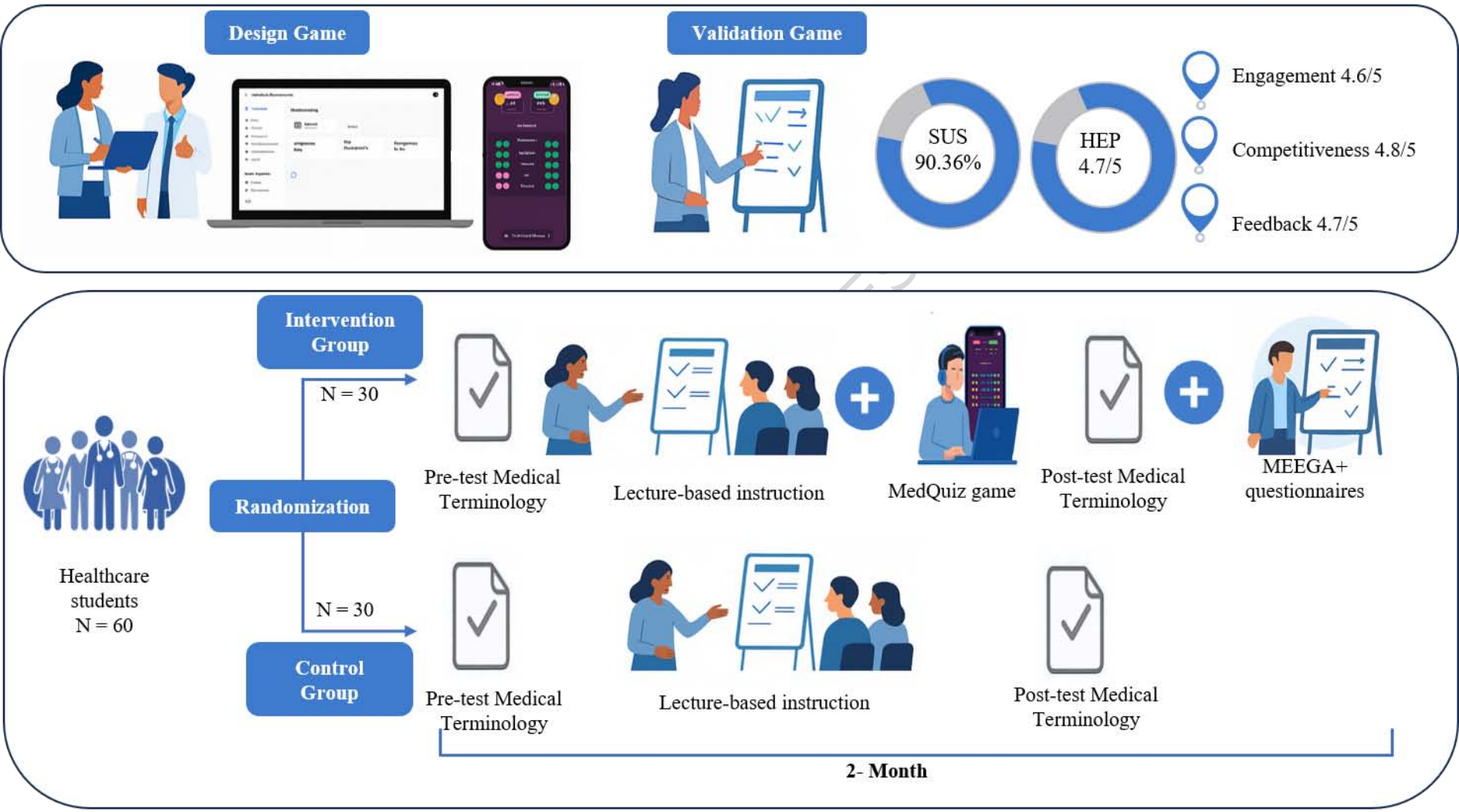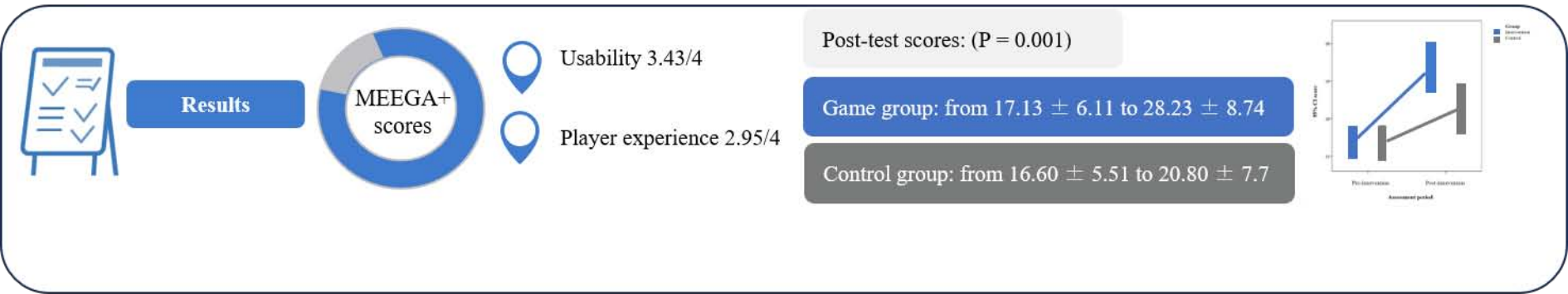
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